

# 100 AWS Interview Questions and Answers for Freshers

Beginner-friendly AWS interview preparation guide covering core cloud, compute, storage, networking, databases, security, serverless, monitoring, containers, and DevOps concepts.

## **1. What is AWS?**

**Answer:** AWS (Amazon Web Services) is a cloud computing platform that provides services for computing, storage, databases, networking, security, and more.

## **2. What is cloud computing?**

**Answer:** Cloud computing means using computing resources such as servers, storage, and databases over the internet instead of maintaining physical infrastructure.

## **3. What are the main benefits of AWS?**

**Answer:** Scalability, flexibility, pay-as-you-go pricing, global infrastructure, high availability, and a wide range of managed services.

## **4. What is an AWS Region?**

**Answer:** A Region is a separate geographical area containing multiple Availability Zones.

## **5. What is an Availability Zone?**

**Answer:** An Availability Zone (AZ) is an isolated location within an AWS Region, designed to improve availability and fault tolerance.

## **6. What is EC2?**

**Answer:** Amazon EC2 (Elastic Compute Cloud) provides resizable virtual servers in the AWS cloud.

## **7. What is an EC2 instance?**

**Answer:** An EC2 instance is a virtual server running in AWS.

## **8. What is an AMI?**

**Answer:** An AMI (Amazon Machine Image) is a template containing the software configuration needed to launch an EC2 instance.

## **9. What is an instance type?**

**Answer:** An instance type defines the compute, memory, storage, and networking capacity of an EC2 instance.

## **10. What is an EBS volume?**

**Answer:** Amazon EBS (Elastic Block Store) provides persistent block storage volumes for EC2 instances.

## **11. What is S3?**

**Answer:** Amazon S3 (Simple Storage Service) is an object storage service used to store and retrieve data.

## **12. What is an S3 bucket?**

**Answer:** An S3 bucket is a container used to store objects such as files and application data.

## **13. What is an S3 object?**

**Answer:** An S3 object is a file or piece of data stored inside an S3 bucket, together with metadata and a key.

## **14. What is S3 versioning?**

**Answer:** S3 Versioning keeps multiple versions of an object, helping protect against accidental overwrites or deletions.

**15. What is S3 lifecycle management?**

**Answer:** S3 Lifecycle rules automatically transition or expire objects according to configured rules.

**16. What is IAM?**

**Answer:** IAM (Identity and Access Management) controls authentication and authorization for AWS resources.

**17. What is an IAM user?**

**Answer:** An IAM user is an identity created for a person or application that needs AWS access.

**18. What is an IAM group?**

**Answer:** An IAM group is a collection of IAM users to which common permissions can be assigned.

**19. What is an IAM policy?**

**Answer:** An IAM policy is a document that defines permissions for AWS resources and actions.

**20. What is an IAM role?**

**Answer:** An IAM role is an identity with permissions that can be assumed by trusted users, applications, or AWS services.

**21. What is the principle of least privilege?**

**Answer:** It means giving a user, application, or service only the permissions required to perform its task.

**22. What is MFA in AWS?**

**Answer:** Multi-factor authentication adds an additional verification factor to help protect AWS accounts and identities.

**23. What is a VPC?**

**Answer:** Amazon VPC (Virtual Private Cloud) is a logically isolated network in AWS where resources can be launched.

**24. What is a subnet?**

**Answer:** A subnet is a range of IP addresses within a VPC used to organize and isolate resources.

**25. What is a public subnet?**

**Answer:** A public subnet has a route that can provide access to or from the internet through an Internet Gateway, depending on its configuration.

**26. What is a private subnet?**

**Answer:** A private subnet does not have a direct route to the internet through an Internet Gateway.

**27. What is an Internet Gateway?**

**Answer:** An Internet Gateway allows communication between a VPC and the internet when routing and security rules permit it.

**28. What is a route table?**

**Answer:** A route table contains rules that determine where network traffic from a subnet or gateway is directed.

**29. What is a security group?**

**Answer:** A security group acts as a virtual firewall controlling allowed inbound and outbound traffic for supported resources.

**30. What is a Network ACL?**

**Answer:** A Network ACL is a subnet-level network filter that controls inbound and outbound traffic using ordered rules.

### **31. Security Group vs Network ACL?**

**Answer:** Security Groups are stateful and associated with resources such as network interfaces. Network ACLs are stateless and operate at the subnet level.

### **32. What is RDS?**

**Answer:** Amazon RDS (Relational Database Service) is a managed service for relational databases.

### **33. Which databases can RDS support?**

**Answer:** RDS supports several relational database engines, including Amazon Aurora, PostgreSQL, MySQL, MariaDB, Oracle, and SQL Server, subject to current AWS offerings.

### **34. What is Amazon Aurora?**

**Answer:** Amazon Aurora is a relational database engine compatible with MySQL and PostgreSQL and designed for cloud workloads.

### **35. What is an RDS Multi-AZ deployment?**

**Answer:** Multi-AZ deployment provides a standby database in another Availability Zone to improve availability and failover.

### **36. What is a database read replica?**

**Answer:** A read replica is a copy of a database used primarily to scale read workloads.

### **37. What is DynamoDB?**

**Answer:** Amazon DynamoDB is a fully managed NoSQL database service designed for low-latency, scalable applications.

### **38. SQL vs NoSQL?**

**Answer:** SQL databases use structured relational tables and SQL queries. NoSQL databases use models such as key-value or document storage and are designed for flexible, scalable workloads.

### **39. What is Lambda?**

**Answer:** AWS Lambda is a serverless compute service that runs code in response to events without requiring server management.

### **40. What is serverless computing?**

**Answer:** Serverless computing lets developers run applications without managing the underlying servers; AWS handles the infrastructure.

### **41. What is a Lambda function?**

**Answer:** A Lambda function is the unit of code deployed to AWS Lambda and executed in response to events or direct invocations.

### **42. What is an event trigger?**

**Answer:** A trigger is an event source that causes a Lambda function to run, such as an S3 event, API request, or scheduled event.

### **43. What is API Gateway?**

**Answer:** Amazon API Gateway is a managed service for creating, publishing, monitoring, and securing APIs.

### **44. What is CloudWatch?**

**Answer:** Amazon CloudWatch monitors AWS resources and applications using metrics, logs, alarms, and related observability features.

**45. What is CloudTrail?**

**Answer:** AWS CloudTrail records API activity and account actions to help with auditing, governance, and security analysis.

**46. CloudWatch vs CloudTrail?**

**Answer:** CloudWatch focuses on monitoring and observability. CloudTrail focuses on recording AWS API activity and account events.

**47. What is CloudFront?**

**Answer:** Amazon CloudFront is a content delivery network (CDN) that delivers content from locations close to users.

**48. What is a CDN?**

**Answer:** A Content Delivery Network distributes cached content through geographically distributed edge locations to reduce latency.

**49. What is Elastic Load Balancing?**

**Answer:** Elastic Load Balancing distributes incoming traffic across healthy targets such as EC2 instances.

**50. What is an Application Load Balancer?**

**Answer:** An Application Load Balancer operates at the application layer and supports HTTP/HTTPS traffic with routing features.

**51. What is a Network Load Balancer?**

**Answer:** A Network Load Balancer operates at the transport layer and is designed for high-performance TCP, UDP, and related traffic.

**52. What is Auto Scaling?**

**Answer:** Auto Scaling automatically adjusts compute capacity according to demand or configured scaling policies.

**53. What is an Auto Scaling Group?**

**Answer:** An Auto Scaling Group manages a group of EC2 instances and can maintain desired capacity and replace unhealthy instances.

**54. What is Elastic Beanstalk?**

**Answer:** AWS Elastic Beanstalk is a managed platform service that simplifies deploying and operating applications.

**55. What is ECS?**

**Answer:** Amazon ECS (Elastic Container Service) is a managed container orchestration service.

**56. What is EKS?**

**Answer:** Amazon EKS (Elastic Kubernetes Service) is a managed Kubernetes service.

**57. What is a container?**

**Answer:** A container packages application code and its dependencies so it can run consistently across environments.

**58. What is ECR?**

**Answer:** Amazon ECR (Elastic Container Registry) is a managed registry for storing and managing container images.

**59. What is Fargate?**

**Answer:** AWS Fargate is a serverless compute engine for containers, commonly used with ECS and EKS.

## **60. What is SNS?**

**Answer:** Amazon SNS (Simple Notification Service) is a managed publish/subscribe and messaging service.

## **61. What is SQS?**

**Answer:** Amazon SQS (Simple Queue Service) is a managed message queuing service used to decouple application components.

## **62. SNS vs SQS?**

**Answer:** SNS primarily distributes messages to subscribers. SQS stores messages in queues so consumers can process them asynchronously.

## **63. What is EventBridge?**

**Answer:** Amazon EventBridge is a serverless event bus service used to route events between applications and AWS services.

## **64. What is Step Functions?**

**Answer:** AWS Step Functions is a workflow service used to coordinate multiple steps and services in an application.

## **65. What is KMS?**

**Answer:** AWS Key Management Service (KMS) helps create and control cryptographic keys used to protect data.

## **66. What is Secrets Manager?**

**Answer:** AWS Secrets Manager securely stores and manages secrets such as database credentials and API keys.

## **67. What is WAF?**

**Answer:** AWS WAF (Web Application Firewall) helps protect web applications from common web exploits and unwanted traffic patterns.

## **68. What is Shield?**

**Answer:** AWS Shield is a managed service that provides protection against Distributed Denial of Service (DDoS) attacks.

## **69. What is AWS Backup?**

**Answer:** AWS Backup is a managed service for centrally automating and managing backups across supported AWS services.

## **70. What is a snapshot?**

**Answer:** A snapshot is a point-in-time backup of supported AWS resources, such as EBS volumes.

## **71. What is CloudFormation?**

**Answer:** AWS CloudFormation lets you define and provision AWS infrastructure using templates.

## **72. What is Infrastructure as Code?**

**Answer:** Infrastructure as Code (IaC) means defining infrastructure in files or code so it can be consistently created and managed.

## **73. What is AWS CLI?**

**Answer:** AWS Command Line Interface is a tool for managing AWS services through command-line commands.

## **74. What is an SDK?**

**Answer:** An AWS SDK provides programming-language libraries that help applications interact with AWS services.

## **75. What is the AWS Management Console?**

**Answer:** It is a web-based interface used to create, configure, monitor, and manage AWS resources.

**76. What is AWS Free Tier?**

**Answer:** AWS Free Tier provides eligible new customers with specified free usage for certain AWS services, subject to current terms and limits.

**77. What is pay-as-you-go pricing?**

**Answer:** It means you generally pay for the AWS resources and services you use, according to the applicable pricing model.

**78. What is scalability?**

**Answer:** Scalability is the ability of a system to handle increased workload by adding or adjusting resources.

**79. Vertical vs horizontal scaling?**

**Answer:** Vertical scaling increases the capacity of a resource. Horizontal scaling adds more resource instances to distribute workload.

**80. What is high availability?**

**Answer:** High availability means designing systems to remain operational with minimal interruption, often by using redundancy.

**81. What is fault tolerance?**

**Answer:** Fault tolerance is the ability of a system to continue operating despite failures of individual components.

**82. What is disaster recovery?**

**Answer:** Disaster recovery is the process and architecture used to restore applications and data after a major failure or disruption.

**83. What is a backup?**

**Answer:** A backup is a separate copy of data that can be used for recovery after accidental deletion, corruption, or failure.

**84. What is a NAT Gateway?**

**Answer:** A NAT Gateway allows resources in private subnets to make outbound connections to the internet while preventing unsolicited inbound internet connections.

**85. What is an Elastic IP?**

**Answer:** An Elastic IP address is a static public IPv4 address that can be associated with an AWS resource, subject to AWS rules and charges.

**86. What is Route 53?**

**Answer:** Amazon Route 53 is a scalable DNS and domain management service with routing and health-check capabilities.

**87. What is DNS?**

**Answer:** DNS translates domain names such as example.com into IP addresses or other DNS records.

**88. What is a hosted zone in Route 53?**

**Answer:** A hosted zone is a container for records that define how traffic should be routed for a domain or subdomain.

**89. What is an Availability Zone used for?**

**Answer:** Availability Zones are used to place resources across isolated locations within a Region to improve resilience and availability.

**90. What is an AWS account?**

**Answer:** An AWS account is a container for AWS resources, billing, security controls, and service usage.

**91. What is AWS Organizations?**

**Answer:** AWS Organizations helps centrally manage multiple AWS accounts with policies, governance, and consolidated billing features.

**92. What is consolidated billing?**

**Answer:** Consolidated billing combines billing information for multiple AWS accounts under an organization.

**93. What is tagging in AWS?**

**Answer:** Tags are key-value labels used to organize, identify, manage, and track AWS resources.

**94. Why is monitoring important in AWS?**

**Answer:** Monitoring helps detect performance problems, failures, unusual activity, and resource usage so issues can be addressed quickly.

**95. What is a deployment?**

**Answer:** Deployment is the process of releasing an application or infrastructure configuration into an environment.

**96. What is CI/CD?**

**Answer:** CI/CD means Continuous Integration and Continuous Delivery/Deployment, practices that automate building, testing, and releasing software.

**97. What is CodeCommit?**

**Answer:** AWS CodeCommit is a source control service; however, AWS has announced its closure to new customers, so GitHub or other current source-control platforms may be used instead.

**98. What is CodeBuild?**

**Answer:** AWS CodeBuild is a managed build service that compiles source code, runs tests, and produces deployable artifacts.

**99. What is CodeDeploy?**

**Answer:** AWS CodeDeploy automates application deployments to supported compute environments.

**100. What is CodePipeline?**

**Answer:** AWS CodePipeline is a continuous delivery service that automates stages such as source, build, test, and deployment.